Bihar Engineering University, Patna

B.Tech 7th Semester Examination, 2025

Course: B.Tech | *Time: 03 Hours*
Code: 100708 | *Subject: Biology for Engineers* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct option / answer the following (Any seven question only):*** **[2 x 7 = 14]**

(a) Which major factor differentiates DNA from RNA?
(i) Guanine (ii) Cytosine
(iii) Thymine (iv) Uracil

(b) Branch of science dealing with classification of organisms is called as _____
(i) Bioautonomy (ii) Taxonomy
(iii) Nomenclature (iv) Specialization

(c) What is the nature of charges that amino acid side chains carry?
(i) Positive (ii) Negative
(iii) Neutral (iv) A and B

(d) Which of the following is not an exclusive characteristic of living organisms?
(i) Consciousness (ii) Movement
(iii) Metabolism (iv) Reproduction

(e) The protein may consist of_____
(i) Polar residues (ii) Non polar residues
(iii) Slightly polar residues (iv) Polar residues and non polar residues

(f) Who is known as the "Father of Genetics"?
(i) Morgan (ii) Mendel
(iii) Watson (iv) Bateson

(g) Which part of the body is the control center for the nervous system?
(i) Leg (ii) Stomach
(iii) Brain (iv) Heart

(h) What is the function of smooth ER (SER)?
(i) Protein production (ii) Provides smoothness to the cell
(iii) Provides attachment with mitochondria (iv) Synthesis and transport of lipids

(i) The simplest amino acid is
(i) glycine (ii) alanine
(iii) asparagine (iv) tyrosine

(j) What is the nature of charges that amino acid side chains carry?
(i) Positive (ii) Negative
(iii) Neutral (iv) A and B

Q.2 (a) Compare Prokaryotic and Eukaryotic cell with suitable diagram. **[7]**
(b) How autotrophs utilize carbon and energy **[7]**

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Q.3	(a)	What are enzymes and explain its role in plants?	[7]
	(b)	What are carbohydrates? Write any four functions of carbohydrates.	[7]
Q.4	(a)	Write short notes on fundamental differences between science and engineering.	[7]
	(b)	Compare Eye and Camera.	[7]
Q.5	(a)	What is Mitotic Cell division? Explain Mitosis with neat diagram.	[7]
	(b)	Give brief account on Dominant Epistasis With suitable example.	[7]
Q.6	(a)	Describe glycolysis in detail with suitable diagram	[7]
	(b)	Explain identification and classification of microorganisms.	[7]
Q.7	(a)	What is the concept of single celled organisms.	[7]
	(b)	Explain briefly the sterilization and media compositions.	[7]
Q.8	(a)	Compare the exothermic and endothermic reactions.	[7]
	(b)	Discuss "Proteins as enzymes, transporters and structural elements".	[7]

Q.9 Write Short notes on ***any two*** of the following: **[7 x 2 = 14]**

(a) Glycolysis
(b) Working of Human Eye
(c) DNA and RNA
(d) Photosynthesis

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